ELI — Memory

ELI v2.3.14

August 2026

Contents

ELI Memory Subsystem

`eli/memory/` — 7.0k LOC, 13 files. The persistent substrate: relational + full-text + vector + graph, all local SQLite/FAISS. Companion to `project_overview.md`.

Files

File	LOC	Role
`memory.py`	4.7k	the `Memory` god-class + `DBPaths` + module facade
`knowledge_graph.py`	643	entity/relation graph (KG)
`habits_memory_db.py`	470	
`vector_store.py`	430	FAISS vector index + embedder
`__init__.py`	0	
`system_index.py`	278	indexed apps/executables/files
`memory_truth.py`	190	
`memory_adapter.py`	131	compat adapter
`memory_service.py`, `sqlite_memory.py`, `stores.py`, `populate_memories.py`	small	helpers/compat

The `Memory` class (`memory.py`)

A single metaclass-backed class (~50 public methods) that owns essentially every persistent concern:

- **Semantic memory**: store_memory, add_memory, recall_memory, search_memory(ies), get_recent_semantic_memori
  adjust_weight, apply_weight_decay.
- **Conversation**: add_conversation_turn, store_conversation, get_conversation_history, get_recent_conversatio
  get_recent_turns_since, search_conversations, get_turns_for_day, save_session_summary,
  get_session_summaries.
- **Habits**: log_habit_event, get_habit_events, add_habit_rule, get_habit_rules, record_habit_run.
- **Self-improvement / learning**: log_learning_event, log_failure, log_correction, add_observation,
  log_improvement, add_capability_proposal, propose_capability, get_pending_proposals, get_recent_failures/im
- **Episodic/semantic/reflective aliases**: store_episodic, store_semantic, recall_semantic,
  store_reflective.
- **Stats / routing**: get_stats, get_dashboard_counts, get_db_routing_info.

vector_store is a lazy property; the KG is integrated via knowledge_graph.get_knowledge_graph().

Schema (≈25 tables)

memories (+ legacy memory), conversation_turns (+ conversations), session_summaries, kg_entities, kg_relations, habit_rules, habit_events, habits, failures, corrections, improvements, observations, capability_proposals, learning_replay, user_patterns, desktop_apps, executables, recent_files, user_dirs, error_tracking, recall_log, events, semantic, emotion_events. FTS5 virtual tables back conversation and KG search.

emotion_events (v2.1.31) — the emotional timeline

Owned by cognition/emotion_timeline.py, not the Memory class: it opens user.sqlite3 directly (same pattern as habits_memory_db.py) and creates its table idempotently, so it adds nothing to the 4.7k-line god-class.

column	meaning
ts, user_id, session_id	when / whose / which session
detected	what the USER seemed to feel
expressed	the register ELI answered in
valence	negative / positive / neutral — what makes "has been negative for a while" answerable without caring which specific emotion each read was
confidence, source, arousal	how strong the read was and where it came from (voice / text / fused / override)
user_text	the utterance that produced the read
eli_prior_action	**the action ELI ran on the PRECEDING turn** — this is the column that lets ELI ask whether the mood turned because of something *it* did

Indexed on ts and (user_id, ts). Reads come back ORDER BY ts DESC, id DESC — the id tiebreak matters because several reads can land inside one second and run-length counting would otherwise mis-order. See blueprints/perception.md for the assessment gates and the proactive surfacing path.

recall_memory — the hybrid retriever (memory.py:1722)

The shared retrieval foundation (see orchestration_and_agents.md for the two strategies on top):

1. **FAISS first** (Stage 5 vector primary). FTS5/LIKE runs only as a *supplement* when the vector index is empty/cold or returns < limit//2 hits. keyword_only=True skips FAISS entirely (the orchestrator runs its own semantic_search, so running FAISS here would double-search with a mislabeled source).

2. **Noise filtering** (important): excludes `assistant_insight`/`episodic`/ `reflection` kinds, `orchestrator` source, and `reflection`/`assistant_insight`/ `session_summary` tags, and rows longer than 1500 chars — so ELI's own old responses/reflections never resurface as "recalled user memories". This was the fix for the "Immutable Techniques" contamination class of bug.
3. **Importance-weighted ordering** via `COALESCE(importance, 0.5)`.

Heavy inline column-detection (`_memory_table_columns`) guards against schema drift across versions — defensive, but a sign the schema has churned.

Vector store (`vector_store.py`)

FAISS `IndexFlat`, embeddings via a local nomic embedder (llama_cpp). Notable: - `_embed_lock` (RLock) serializes embedding — the embedder is **not thread-safe** and concurrent calls segfault (this is why the orchestrator retrieval is sequential). - Metadata canonicalized to **`meta.json`** (migrated from legacy `meta.pkl`). - Singleton via `get_vector_store()`; shutdown-aware (skips embedding during teardown); `reset_vector_store()` for rebuilds.

Knowledge graph (`knowledge_graph.py`)

`kg_entities(name,type,aliases,description,confidence)` + `kg_relations(subject_id, predicate, object_id, weight, source)` — a subject-predicate-object graph. FTS5 over entities (with insert/update/delete triggers) for fuzzy `search_entities`. `upsert_entity`, `context_for_prompt` (lightweight SQLite-only prompt context — no embedding). Stop-word list prevents common words becoming entities.

Truth layer (`memory_truth.py`)

`inspect_sqlite` / `inspect_vector_store` — read-only authority/inspection used by status surfaces; reads `vectors/meta.json` preferentially, falls back to legacy pickle. Backs `truth_report` and the memory-status surfaces.

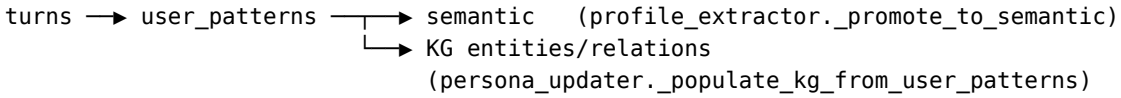
Weight decay & consolidation

`apply_weight_decay(min_weight=0.05, older_than_days=7, half_life_days=30)` — recomputes `weight` as `2 ** (-age_days / half_life)`, with the half-life stretched by importance (`DECAY_IMPORTANCE_STRETCH`) and rows at or above `DECAY_PIN_IMPORTANCE` (0.85) pinned at 1.0. Because it is a pure function of age and importance it is idempotent — its caller samples ~1% of responses, and a missed tick cannot leave a memory over-weighted.

It previously multiplied the stored weight by a constant on each call, which measured how often the function ran rather than how old a memory was. On a live 441-memory store every row was still at exactly 1.0, and since `weight` carries 15% of the recall fusion score (`scoring.RERANK_W_WEIGHT`) that term was a constant contributing no ranking signal at all.

`consolidate_memories(dry_run=False)` — folds exact-duplicate memory text down to one canonical row per group, keeping the group's best importance, newest timestamp and unioned tags. Reflection's dedupe guard was fixed to do an existence check rather than a relevance query, but nothing retired what had already accumulated: the same live store held 169 exact-text duplicates across 66 groups — 38% of everything — one insight repeated 28 times. It also rebuilds `memories_fts` after deleting, because **the FTS triggers declared in the schema do not exist on any database, fresh or live** — the index is maintained by an explicit INSERT in the store path and nothing had ever deleted from it. Note that orphaned index entries cannot be detected from SQL: on an external-content FTS5 table any query without a `MATCH` reads through to the content table, so `COUNT(*)` and `SELECT rowid` only ever report live rows.

Promotion across tiers is automatic, not manual. It is a fan-out from `user_patterns` rather than the linear chain the name suggests:



`_promote_to_semantic` applies a durable/transient prefix filter, dedupes against the `semantic` table, and only fires when `_insert_user_pattern` actually inserted, so reaffirmations do not re-promote. `_populate_kg_from_user_patterns` maps six pattern prefixes to typed relations off the `User` node. `store_episodic`/`store_semantic` are indeed thin aliases, but they are not the promotion path and never were.

The real weakness is the **width of the funnel, not its absence**: 619 turns and 441 memories yielded only 10 `user_patterns`, and everything downstream inherits that — 6 semantic facts, 27 KG entities. Extraction is regex-driven, and three of those ten patterns are `app_cmd` JSON blobs the KG mapper discards.

Honest assessment

- **Strong:** genuinely hybrid (vector + FTS5 + KG) on one local SQLite foundation; the noise-filtering in `recall_memory` is the right instinct and fixed real contamination; embedder serialization correctly avoids the segfault.
- **Weak:**
  1. `memory.py` is a **4.5k-line god-class** spanning ~8 unrelated concerns (semantic, conversation, habits, learning, failures, capabilities, system index). Wants to be split along those seams.
  2. **Schema sprawl / redundancy** — `memories` *and* `memory`, `conversations` *and* `conversation_turns`, plus a standalone `semantic` table. The inline column-detection everywhere is compensating for schema instability.
  3. **No consolidation pipeline** — only multiplicative decay; memories don't graduate into the KG automatically.

Update — 2026-06-09

- **Detected habits readable** (`Memory.get_detected_habits(min_count, limit)`): the proactive daemon fills the `habits` table via `detect_habits()`, but HABIT_STATUS, the persona overlay, and the bus `HabitAgent` all previously read only the (usually empty) `habit_rules` table → "no habits detected". `get_detected_habits` reads the real `habits` table with a meta/introspection denylist (filters SELF_REPORT/MEMORY_STATUS/EXPLAIN_* noise from the user testing ELI), so genuine behaviour (media, app launches, screenshots) surfaces.
- **FAISS persistence bug fixed:** `_eli_persist_loaded_vector_store` was writing vector metadata with `pickle.dump` to `meta.json` while `_load_meta` reads JSON — so a rebuild corrupted the index and the next load silently auto-rebuilt (accumulating phantom vectors). Both write sites now use the canonical `_dump_meta` (JSON); the index re-syncs cleanly to the memory count. (This also closes a pickle-RCE-on-load vector the codebase had deliberately retired.) KG populator enriched (broader, clean entity extraction from `user_patterns`).

Update — 2.3.7 (recent-history window widened)

`runtime/memory_evidence.py` capped every recent-history pull at `max(4, min(limit, 8))`. The cap **silently ignored a larger limit**: a caller asking for 40 recent turns still received 8, so continuity was being thrown away by a constant nobody could see or configure.

- `RECENT_HISTORY_CAP = 40` — the ceiling now follows the caller, bounded only by a sane upper limit so a huge `limit` cannot drag the whole conversation table into a single turn.
- `collect_memory_evidence(limit=32)` and `build_memory_evidence_text(limit=32)` — defaults raised from 12 and 8.

This is separate from `cog.mem_recent_turns` (default 24, max 80), the user-facing tunable in Settings ▸ Cognition that governs how many recent turns enter the prompt. The prompt assembler still does

the real budgeting downstream; this change only stops the evidence layer discarding history before the budgeter ever sees it.